

BY: Mahmood

(Mth104) Set and logic

Lecture 23

Ordering of Cardinal Numbers

• Real numbers and power of Continuum:

Theorem 1

The unit interval $I = [0, 1]$ is noncountable.

• Proof:

let I is countable then

$$I = \{x_1, x_2, x_3, \dots\}$$

So,

$$x_1 = 0.a_{11}, a_{12}, a_{13}, \dots, a_{1n}$$

$$x_2 = 0.a_{21}, a_{22}, a_{23}, \dots, a_{2n}$$

$$x_3 = 0.a_{31}, a_{32}, a_{33}, \dots, a_{3n}$$

$$= \{0.1, 2, \dots, 9\}$$

$$x_1 = 0.000, 0.009, 0$$

$$x_2 = 0.4999$$

$$x_3$$

$$0 = 0.000\dots$$

$$0.1 = 0.0999\dots$$

$$0.5 = 0.4999\dots$$

$$1 = 0.999\dots$$

let construct the new real number.

$y = 0.b_1 b_2 b_3 \dots b_n$
which belong I in way that.

$$b_1 \neq a_1, \quad b_1 \neq 0$$

$$b_2 \neq a_2, \quad b_2 \neq 0$$

$$\text{therefore } b_3 \neq a_3, \quad b_3 \neq 0$$

$$y \neq x$$

Hence

I is non denumerable.

Power of Continuum: "size of non denumerable"

if set A has the same cardinality
as the unit interval $I = [0, 1]$ then
it is said to have power of continuum

Example:

$$[a, b] \approx [0, 1]$$

close interval $[a, b]$, $[a, b)$, $(a, b]$ and (a, b)
have power of continuum.

Proof,

the unit interval is non denumerable and
has same cardinal as set of real num which
have power of continuum.

• Cardinal Numbers:

Each set A is assigned a symbol in such that two sets A and B are assigned the same symbol if they are equipotent.
denoted by: $|A|$, $n(A)$ or $\text{card}(A)$

$$|A| = |B| \quad \text{iff} \quad A \approx B$$

• Finite Cardinal Number:

cardinality of empty set $\emptyset$ is "0" and cardinality of set $\{1, 2, 3, \dots, n\}$ is " n ".

Example:

$$|A| = 3 \quad \text{iff} \quad A \approx \{1, 2, 3\}$$

$$|A| = 5 \quad \text{iff} \quad A \approx \{1, 2, 3, 4, 5\}$$

OR.

$$\textcircled{1} \quad S = \{4, 3, 2\}$$

What is the cardinality of S

$$|S| = 3$$

$$\textcircled{2} \quad B = \{\emptyset\} = \{1\} \quad |B| = 1$$

infinite Cardinal Numbers:

Cardinal Number of infinite set called infinite or transfinite cardinal numbers.
Infinite Set:

① Positive integers.

the cardinal num of infinite positive integer is $\aleph_0$ (aleph nought) $\aleph_0$.

$$|A| = \aleph_0 \text{ iff } A \approx \mathbb{P}.$$

② $|\mathbb{Z}| = \aleph_0$ since $\mathbb{Z} \approx \mathbb{P}$

③ $|\mathbb{Q}| = \aleph_0$ since $\mathbb{Q} \approx \mathbb{P}$

④ Unit interval $I = [0, 1]$

the cardinal num of unit interval is denoted by c .

$$|A| = c \text{ iff } A \approx I$$

$$|R| = c \text{ since } R \approx I$$

$$|(a, b)| = c \text{ since interval } \approx I$$

• denumerable = $\aleph_0$

• non denumerable = c

• Ordering of sub Cardinal number:-

let A and B be two set
We say that $|A| \leq |B|$ if there exist
one-to-one function



• Example:

let n be a finite cardinal $|A| < |B|$

let A be a finite subset of $P = \{1, 2, 3, \dots\}$
then, then $|A| \neq |P|$

Thus

$$|A| < |P|$$

$$n < \aleph_0$$

$$0 < 1 < 2 < 3 < \dots < \aleph_0$$

$$|P| \leq I \quad \text{or} \quad |P| = |I|$$

$$|P| < |I|$$

$$\aleph_0 < C$$

$|P| \neq |I|$ coz there is

no way to put them into
one-to-one correspondence

$$\aleph_0 < \aleph_1 < C$$

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Lecture 24

Cantor's theorem

For any Set A , we have $|A| < |P(A)|$

• proof:

Let $f: A \rightarrow P(A)$

let $x \in A \Rightarrow f(x) \in P(A)$

$$= f(x) \subseteq A$$

$$x \in A, f(x) \subseteq A$$

either

$$x \in f(x) \text{ or } x \notin f(x)$$

$$\text{let } X = \{x; x \in A, x \notin f(x)\}$$

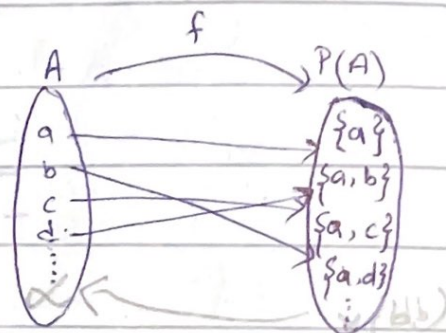
$$X \subseteq A + x \in P(A)$$

$\therefore f$ is onto mapping

then Pre image exist.

$$\text{let } \alpha \in A$$

$$f(\alpha) = X$$



$$f(a) = \{a\} \Rightarrow a \in f(a)$$

$$f(b) = \{a, d\} \Rightarrow b \notin f(b)$$

$$f(c) = \{a, c\} \Rightarrow c \in f(c)$$

$$f(d) = \{a, b\} \Rightarrow d \notin f(d)$$

$$f(\alpha) = \{x\} \Rightarrow \alpha \notin f(\alpha)$$

Set X = set of those element

which does not belong

to function $X = \{b, d\} \Rightarrow X \subseteq A$

• let onto mapping, pre image $X \in P(A)$

$$f(\alpha) = X$$

$$\alpha \in X \text{ or } \alpha \notin X$$

$$\alpha \in f(\alpha) \mid \alpha \notin f(\alpha)$$

∴ Pre image $\alpha \in A$

$$f(\alpha) = X \quad \text{--- (1)}$$

either

$$\alpha \in X \quad \text{or} \quad \alpha \notin X$$

$$\alpha \in f(\alpha) \quad \text{or} \quad \alpha \notin f(\alpha)$$

$$\alpha \notin X \quad \text{or} \quad \alpha \in X$$

Let onto mapping

Pre image of x exist

$$f(\alpha) = X \quad \text{--- (1)}$$

$$\text{either } \alpha \in X \quad \text{or} \quad \alpha \notin X$$

$$\Rightarrow \alpha \in f(\alpha) \quad \text{or} \quad \alpha \notin f(\alpha)$$

$$\alpha \notin X \quad \text{or} \quad \alpha \in X$$

Hence it is not possible, f can not be onto mapping.

• if A has 3 element then
 $P(A) = 2^3 = 8$

$$|P| < |P(P)|$$

$$N_0 < 2^{N_0}$$

$$|R| < |P(R)|$$

$$C < 2^C$$

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Lecture 25

Cardinal Arithmetic

Addition and multiplication of counting numbers from the point of set theory

e.g. $2+3=5$

$$A = \{1, 2\} \quad B = \{3, 4, 5\}$$

← disjoint set

$$A \cup B = \{1, 2, 3, 4, 5\}$$

(5)

• Cardinal Addition:-

let α and β be cardinal numbers and let A and B be disjoint set with $\alpha = |A|$ and $\beta = |B|$ then the sum of α and β is =

$$\alpha + \beta = |A| + |B|$$

Example:

$$A = \{1, 2, 3, \dots, n\} \Rightarrow |A| = n$$

$$B = \{n+1, n+2, n+3, \dots\} \Rightarrow |B| = \aleph_0$$

$$A \cup B = \{1, 2, 3, \dots, n, n+1, n+2, n+3, \dots\} = |A \cup B| = \aleph_0$$

$$\text{But } |A \cup B| = n + \aleph_0$$

$$\aleph_0 = n + \aleph_0$$

} $B \in P$

$$E = [0, \frac{1}{2}]$$

$$F = (\frac{1}{2}, 1]$$

$$E = [0, \frac{1}{2}] \Rightarrow |E| = c$$

$$F = (\frac{1}{2}, 1] \Rightarrow |F| = c$$

$$\left\{ \begin{array}{l} E \cup I \Rightarrow [0, 1] \\ \downarrow \\ c \end{array} \right.$$

$$E \cup F = [0, \frac{1}{2}] \cup (\frac{1}{2}, 1] = [0, 1] = c$$

$$|E \cup F| = |E| + |F|$$

$$c = c + c$$

Cardinal Multiplication:-

Let α and B be cardinal num let A and B be set with $\alpha = |A|$ & $B = |B|$ then $\alpha \times B = |A \times B|$

Example:

$$A = \{2, 3, 4\} \quad B = \{4, 5\}$$

$$|A| = 3$$

$$|B| = 2$$

$$|A \times B| = 6 = 3 \times 2 = |A| \times |B| = \alpha B$$

Theorem 1.

Let α and B non zero cardinal number such that B is infinite and $\alpha \leq B$ then

$$\alpha + B = \alpha B = B$$

$$\left\{ \begin{array}{l} \alpha + B = B \\ \alpha B = B \end{array} \right.$$

We know that

$$N_0 \leq c$$

Positive num $<$ Real num

$$N_0 + c = c$$

$$N_0 c = c$$

Ordered Sets

if a relation on set satisfying,
 reflexive, antisymmetric and transitive property
 then the relation R is called order relation
 or partial order.

- reflexive $\rightarrow$ for any set, we have aRa . $\{1, 2, 3\}$
 $(1,1) (2,2) (3,3)$
- antisymmetric $\rightarrow$ for any set, if aRb & bRa then $a=b$
- transitive $\rightarrow$ if aRb and bRc then aRc .

$(1,2) (2,1) (1,1)$

Example:

Consider the set P of positive
 integer. We say a divided b written
 $a|b$ if there exist an integer c .

$a|b$

- 1) If $a \in P$, $a|a$ or aRa . ✓
- 2) if $a|b$ and $b|a$ then $a=b$. ✓
- 3) if $a|b$ and $b|c$ then $a|c$. ✓

the relation " $|$ " divisibility is not an ordering of set $\mathbb{Z}$ integer.

- 1) $a|a, aRa = -2|-2, -2R-2$
- 2) for example $2|-2$ and $-2|2$
but $2 \nmid 2$ ✗

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Lecture 27

Comparability and Linearly Ordered Sets

• Comparability:

Comparability is property of element in partial order set. In partial order set two element are comparable

if $a \leq b$, $b \leq a$

(a, b) (b, a) $(1, 2)$

• Non-Comparable:

If two element are not comparable it mean that they are unrelated. There is no clear ordering between them.

$(3, 4)$

$(1, 2)$

$a \parallel b$ if $a \not\leq b$ or $b \not\leq a$

• Linearly Ordered Set and chain:

If every pair of element of set are comparable then set is said to be linearly ordered or totally ordered.

set $(1, 2, 3)$

$(1, 2)$ $(2, 3)$

$1 < 2$ $2 < 3$

Example,

- the set of natural num
 $(\{1, 2, 3, \dots\}, \leq)$ with usual ordering ✓
 - $(\{-3, -2, -1, 0, 1, \dots\}, \leq)$ with usual order ✓
 - $(\{1, 2, 3, 4, \dots\}, |)$ order by divisibility $3|4$ ✗
(a, b) $(\{a\}, \{b\})$, order by set inclusion $\subseteq$ ✗
- chain:

linearly ordered subset of order
pair set is called chain.

Example: $\{1, 2, 3, 4, 5, 6, \dots\}$ $\div$

- subset = $\{2, 6, 12, 36\}$ order by $\div$ from
positive integer chain

~ $A = \{1, 2, 3, 4, 6, 12, 24\}$ order by $\div$
Sol:

No linearly order coz 3, 4 are not comp

We observe that, $\{1, 2, 4, 12, 24\}$ } these subset are
 $\{1, 2, 6, 12, 24\}$ } Chain
 $\{1, 3, 6, 12, 24\}$ }

let $P = \{1, 2, 3, \dots\}$ order by divisibility state whether these are chain or not.

$$A = \{24, 2, 6\}$$

$2 \mid 6$ $6 \mid 24$ it is a chain

$$B = \{3, 15, 5\}$$

$3 \nmid 5$ its not a chain

$$C = \{2, 8, 32, 4\}$$

$2 \mid 4 \mid 8 \mid 32$ its a chain

$$D = \{7\}$$

$7 < 7$ is a chain ✓

Difference b/w

Predecessor & Immediate Predecessor

$$Q: \{1, 2, 3, 6, 4, 12, 24\} :$$

$$\{1, 2, 6, 12, 24\}$$

$$\{1, 2, 4, 12, 24\}$$

$$\{1, 3, 6, 12, 24\}$$

$$\cdot \{1 < 2 < 6 < 12 < 24\}$$

$$\{1 < 2 \quad 2 < 6 \quad 6 < 12 \quad 12 < 24\}$$

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Lecture 28

Partially Ordered Set and Hasse Diagram

Immediate Predecessor and Immediate Successor:

- $a \ll b$ $a \leq b$
 - $b \ll a$ $b \leq a$
- } If no element lie b/w a, b lie between a, b

Example:

$$Q_1 \quad P = \{1, 2, 3, 4, \dots\} <$$

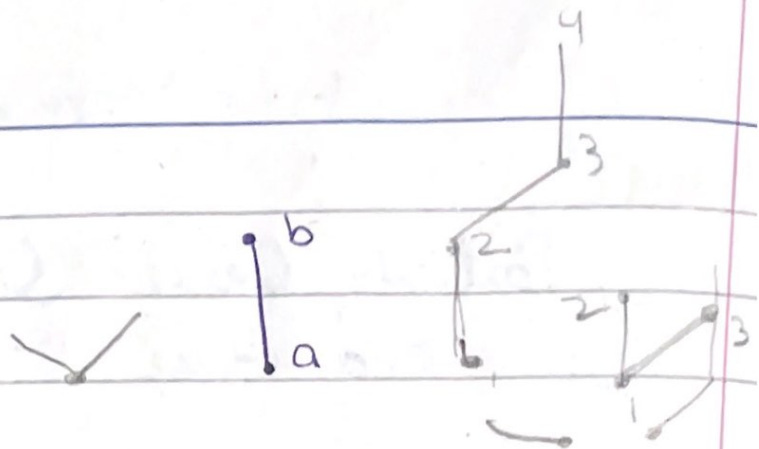
$1 \ll 2, 2 \ll 3, 3 \cdot 4$

$$Q_2 \quad A = \{1, 2, 3, 4, 6, 12, 24\} \div$$

$$2 \ll 4, 2 \ll 6, 3 \ll 6, 4 \ll 12, 6 \ll 12$$
$$12 \ll 24, 4 \ll 12, 6 \ll 12$$

$$\left. \begin{array}{l} \{1 \ll 2 \ll 4 \ll 12 \ll 24\} \\ \{1 \ll 2 \ll 6 \ll 12 \ll 24\} \\ \{1 \ll 3 \ll 6 \ll 12 \ll 24\} \end{array} \right\} \text{Chain}$$

Hasse Diagram:



Example:

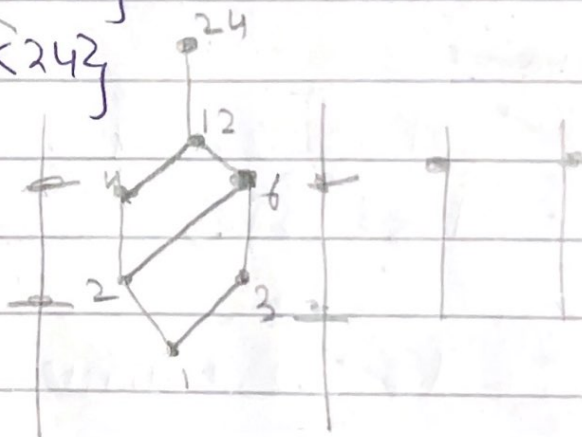
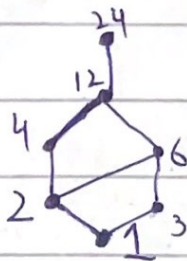
Draw the Hasse diagram

$$A = \{1, 2, 3, 4, 6, 12, 24\}$$

Sol:-

1. $\{1 \ll 2 \ll 4 \ll 12 \ll 24\}$
2. $\{1 \ll 2 \ll 6 \ll 12 \ll 24\}$
3. $\{1 \ll 3 \ll 6 \ll 12 \ll 24\}$

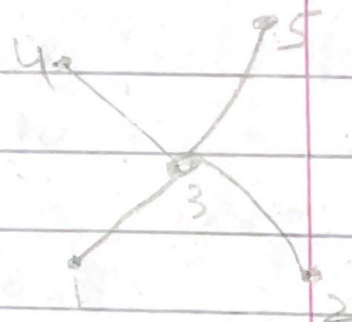
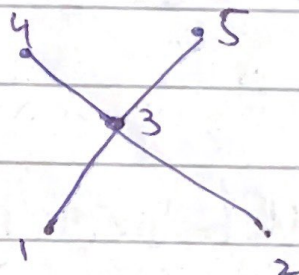
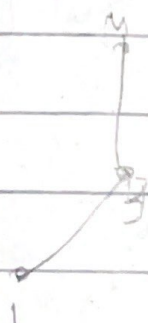
1.



non comparable should be in parallel

$A = \{1, 2, 3, 4, 5\}$ & Relation $R =$
 $\{(1,1)(1,4)(1,5)(2,2)(2,4)(2,3)(2,5)$
 $(3,3)(3,4)(3,5)(4,4)(5,5)\} \setminus \{(1,3)\}$

Sol. $\{(1,3)(1,4)(1,5)(2,3)(2,4)(2,5)$
 $(3,4)(3,5)\}$



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Lecture 29

Minimal and Maximal element.

• Minimal Element

let S be partially ordered set. An element $a \in S$ is called a minimal element of S if no element of S strictly precedes (is less than) a .

Example.

$A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$
ordered by divisibility.

$1 \leq 2, 1 \leq 3, 1 \leq 4, 1 \leq 6, 1 \leq 8, \dots$

• Maximal Element:

let S be partially ordered set. An element $b \in S$ is called maximal element of S if no element of S strictly succeeds (is greater than) b .

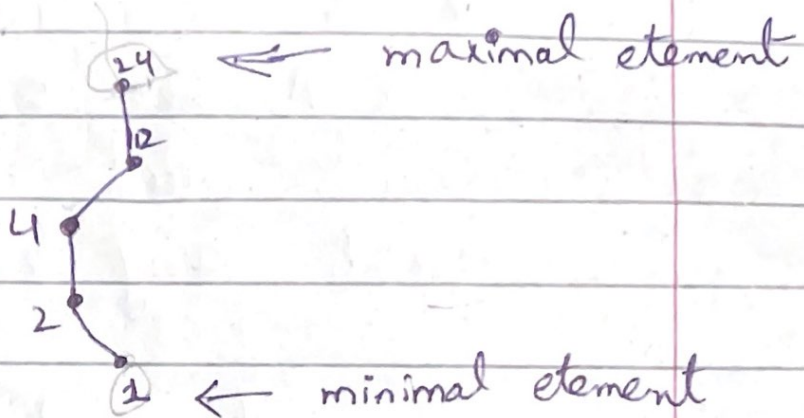
Example:

$$A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$$

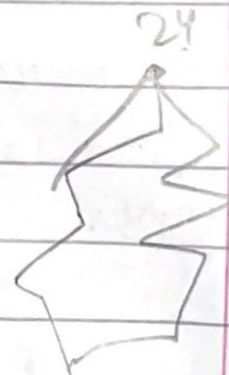
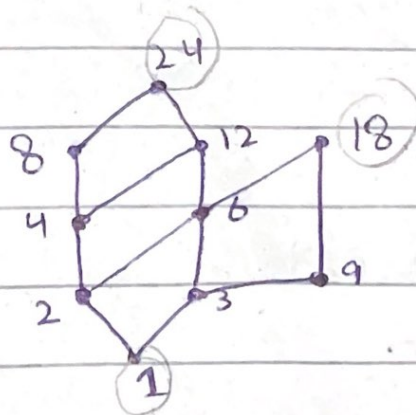
$A = 18, 24$ are two maximal element

Hasse diagram:

$$A = \{1, 2, 4, 12, 24\}$$



$$A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$$



1 = minimal element | First element

24, 18 = maximal element | No last

element coz
it should
succeed every
element in set

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Lecture 30

Supremum and Infimum

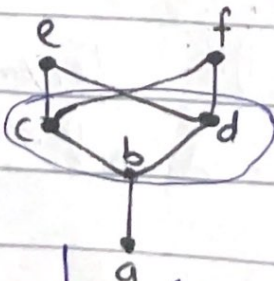
• Upper Bound and Supremum:

let S be partially ordered set, and let A be subset of S .
an element m is called upper bound if it succeeds every element $x \leq m$

- if an upper bound of A precedes every upper bound of A then it is called supremum.

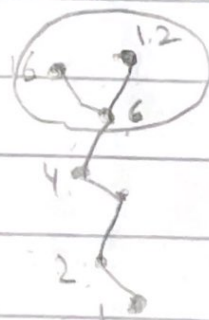
Example:

$S = \{a, b, c, d, e, f\}$
subset $= A = \{b, c, d\}$



Upper bound = e, f

Supremum = e and f are non-comparable hence no supremum exist

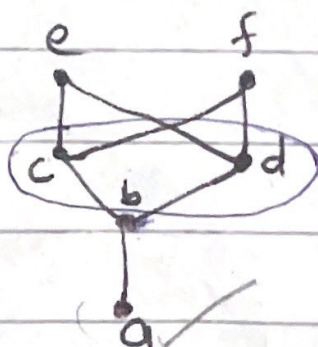


Lower Bound and Infimum:

Let S be partial ordered set and A be a subset of S . An element n in S is called lower bound of A if n precede every element of A : $y \geq n$

- If lower bound of A succeed every other lower bound of A then it is called infimum of A .

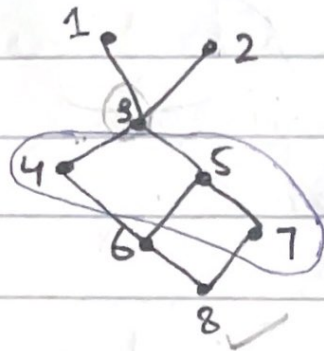
Example:



lower bound = a, b

Infimum = b

let $S = \{1, 2, 3, \dots, 8\}$ be ordered as in the Hasse diagram below let $A = \{4, 5, 7\}$



• Upper bound = 1, 2, 3
Supremum = 3

• lower bound = 8
Infimum = 8

Supremum and Infimum should
or should not be from subset.

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Lecture 31

Similar order Set

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Similar Mapping:

Suppose X and Y are partially ordered sets. A one-to-one injective function $f: X \rightarrow Y$ is called similar mapping.

1. if $a \leq b$ then $f(a) \leq f(b)$
2. if $a \parallel b$ (noncomparable) then $f(a) \parallel f(b)$

Similar ordered-Sets: $X \simeq Y$

It is also called

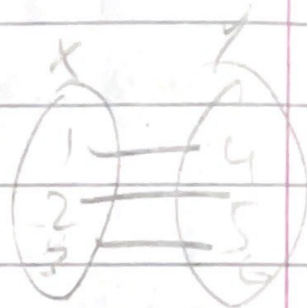
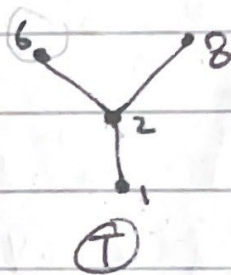
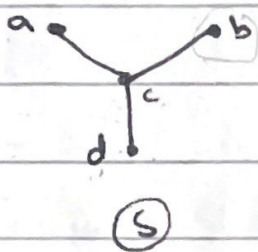
order-isomorphism or isomorphism or partial ordered set or similar etc.

if there exist a ^{one-to-one correspondence} bijective function then it is called

and which preserve the order relation is called isomorphism or order-isomorphism.

Example:-

Suppose $S = \{a, b, c, d\}$ and
Suppose $T = \{1, 2, 6, 8\}$ is order
by divisibility determine whether $S \cong T$?



define function

$$f: S \rightarrow T$$

1. $f(a) = 6, f(b) = 8, f(c) = 2, f(d) = 1$

is

2. order preserve = $d \leq c \Leftrightarrow f(d) \leq f(c)$

$$c \leq b \Leftrightarrow f(c) \leq f(b)$$

$\therefore$ Hence $S \cong T$

Consider the usual ordering $\leq$ of the positive integer $P = \{1, 2, 3, \dots\}$ and negative integer $A = \{-1, -2, -3, \dots\}$

Sol:-

1. let $f: P \rightarrow A$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}$$

2. Order preserve $1 \leq 2 \quad f(1) \leq f(2)$

$$1 \leq 2 \quad -1 \leq -2 \quad (\text{no})$$

Hence P is not isomorphic to A .

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Lecture 32

Well ordered Set and Initial Segment

Well ordered Set,

Let A be an ordered set then A is said to be well-ordered if every subset of A contain a first element.

• In partial ordered there are non comparable and comparable element but in well ordered set must be linearly ordered.

Example:

two subset of well order set P .

$$A_1 = \{1, 3, 5, 7, \dots\}$$

$$A_2 = \{2, 4, 6, 8, \dots\}$$

Both are also well ordered because both has first element.

Initial Segment:

initial segment of element consist of all element which strictly precede a .

Example:

$S = \{1, 3, 5, 7, 9, 11\}$ build
Initial segment of 5 and 9.

$$S(5) = \{1, 3\}$$

$$S(9) = \{1, 3, 5, 7\}$$

$$S(1) = \emptyset$$

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Lecture 33

Transfinite Induction

Limit element, Ordinal number

Principle of transfinite Induction:

Let S be a subset of well ordered set A with following properties $A = \{1, 2, 3, 4, \dots\}$

1) $a_0 \in S$, a_0 is first element of A . $S = \{1, 2, 3\}$

2) if $s(a) \subseteq S$ then $a \in S$
then $S = A$.)

Example:

$$A = \{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$$

$$S = \{1, 2, 3\}$$

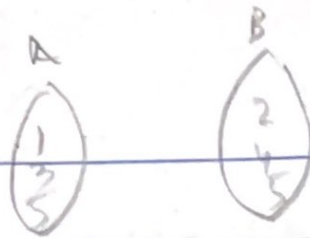
$$a_0 = 1 \Rightarrow \text{first element } A$$

• Limit Element:

An element (a) in a well ordered set A is called limit element

if it does not have an immediate predecessor. and if it is not the first element. $\{1, 2, 3\}$

Example:



$$A_1 = \{1, 3, 5, \dots\} \quad A_2 = \{2, 4, 6, \dots\}$$

Sol:

$$S = A_1 \cup A_2 = \{1, 3, 5, \dots, 2, 4, 6, \dots\} \quad \{1, 2, 3, 4, 5, \dots\}$$

Hence

- ✓ 1 and 2 do not have immediate predecessor.
- 1 is first element and 2 is not first element.
- ∴ Limit element is 2.

• Comparison of well ordered sets:

Suppose A and B are well ordered set, suppose A is similar to initial to an initial segment of B then A is said to be shorter than B.

$$\text{Example: } A = \{a_1, a_2, a_3, a_4\} \quad B = \{b_1, b_2, b_3, b_4, b_5\}$$

$$\text{Now } S(b_5) = \{b_1, b_2, b_3, b_4\}$$

Here we see that $S(b_5) \cong A$ because there exist one-to-one correspondence b/w A and $S(b_5)$

Ordinal number:

- order of element in set.
- collection of well order set

$\text{ord}(A)$

Cardinal number.

- number of element in set.
- collection of set

$\text{card}(A), |A|$

finite Ordinal Number:

$\{1, 2, 3, \dots\}$

$\{\}, \{\}, \{\}, \emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}, \dots$
is denoted by $0, 1, 2, 3, \dots$

- Ordinal number of P : is denoted by ω .

Inequalities and ordinal number:

$$\lambda = \text{ord}(A)$$

$$\mu = \text{ord}(B)$$

If A is similar to initial segment of B

ω $\lambda < \mu$, A is shorter than B ✓

$\lambda = \mu$, A is similar to B .

$\lambda > \mu$, A is longer than B .

Choice function and axiom of choice, Well ordering theorem

Choice function:

Let C be a nonempty set of non-empty sets. $f: C \rightarrow \cup C$ is called choice function for C if $f(A) \in A$ for all $A \in C$.

Example:

$$C = \{ \{1, 2\}, \{3, 4\}, \{5, 6\} \}$$

Rule $\Rightarrow$ choose second element from A

- $f(A) \in A$

$$f(\{1, 2\}) = 2 \quad \text{choose 2 from set } \{1, 2\}$$

- $f(B) \in B$ [Rule choose first element]

$$f(\{3, 4\}) = 3 \quad \text{choose 3 from set } \{3, 4\}$$

- Axiom of Choices

give that there always exist a choice function.

Well ordering theorem:-

Every set can be well-ordered

Although $\mathbb{Z}$ with usual order is not well ordered, so we define another kind of order on it which does make it well order. we give previous order $\leq$ $\leq$ $\leq$

$a < b$ whenever $|a| < |b|$ or $|a| = |b|$

Example ::

$$|-9| = 9$$

$$|-9| < |-20|$$

$$9 < 20$$

means

$$-9 < -20$$

$$|-1| < |-5|$$

$$1 < 5$$

$$-1 < -5$$

$$|-2| = |2|$$

$$2 = 2$$

$$-2 < 2$$

$$|0| < |5|$$

$$0 < 5$$

$$0 < 5$$

$$\mathbb{Z} = \{ \dots, -2, -1, \underline{0}, 1, 2, \dots \}$$

• Zorn's lemma:

$$X = \{1, 2, \dots\}$$

let X be non empty partial order set in which every chain has upper bound x . then X contains at least

one maximal elements. i.e. $X = \{1, 2, 3, 4, 6, 12, 24\}$

$\{1, 2, 4, 12, 24\}$ $\{2, 3, 6, 12, 24\} \Rightarrow 24$ is upper bound Also maximal element

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Lecture 35-36

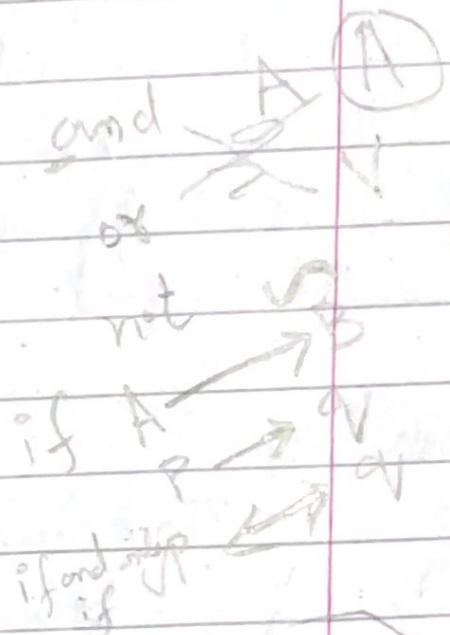
Proposition and Compound proposition Logical Operation

Proposition:

A proposition is a declarative sentence which is true or false.

Example:

- $1+1=2$ T ✓
- $2+2=3$ F
- Islamabad is in Paris F
- Where are you going?
- Do your homework.



Compound proposition:

- when two or more propositions are combined using logical operators.

Example:

- Rose are red and violet are blue
- Ali is intelligent or he studies full night.

not
if
if and only if

and
or
not
conditional
Biconditional

it is not hot

$\neg P$

it is hot and sunny

$P \wedge Q$

it is not hot or sunny

$\neg P \vee Q$

it is not hot but sunny

$\neg P \wedge Q$

it is neither hot nor sunny

$\neg P \wedge \neg Q$

* Negation: ($\neg$) not

$\neg P$

P	$\neg P$
F	T
T	F

$\neg Q$

Q	$\neg Q$
T	F
F	T

* Conjunction: ($\wedge$) And

$P \wedge Q$

$$2^3 = 8$$

$$2^2 = 4$$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

* Disjunction: ($\vee$) or

$P \vee Q$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

* Implication / Conditional Statement: $\rightarrow$

$$P \rightarrow q$$

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

if P then q or P implies q

Examples:-

if	$1=1$	then	$2=3$	True
	$1=1$ T		$2=3$ F	False
	$1=0$ F		$3=3$ T	True
	$1=0$ F		$2=3$ F	True

$$P \rightarrow q \equiv \neg q \rightarrow \neg p$$

P	q	$\neg p$	$\neg q$	$P \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Inverse of Conditional Statement $p \rightarrow q$ is $\neg p \rightarrow \neg q$

* if P is a square, then P is a rectangle

* if P is not a square, then P is not a rectangle

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MTH202 - Discrete Math

Lecture 4

Biconditional law of logic involving
Biconditional and their ApplicationDouble Implication / Biconditional: $p \leftrightarrow q$

if and only if

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

$$\neg p$$

T	F
F	T

$$\neg$$

T	T
T	F
F	T
F	F

$$\vee$$

T	T
T	F
F	T
F	F

$$\rightarrow$$

T	T
T	F
F	T
F	F

Example:

1) $1+1=3$ if and only if earth is flat = True
 $F \leftrightarrow F = T$

2) Sky is blue iff $1=0$ = false
 $T \leftrightarrow F = F$

3) Milk is white iff birds lay egg = True
 $T \leftrightarrow T = T$

4) $x > 5$ iff $x^2 > 25$ False
 $T \leftrightarrow T = T$

$x = 6$
 $6 > 5$ is True
 $6^2 = 36 > 25$ is True
 $T \leftrightarrow T = T$

$x = 7$
 $7 > 5$ is True
 $7^2 = 49 > 25$ is True
 $T \leftrightarrow T = T$

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Lecture 37-38

Real Number and power of Continuum part 1, part 2

Real number and power of continuum }
theorem 1 $\leftarrow$ part 1 ✓
theorem 2 $\leftarrow$ part 2 ✓
Cardinal number $\leftarrow$ part (38 lecture) ✓
finite cardinal number
infinite cardinal number

} Lecture 23

(Mth 104)

Lecture 39

Tautology and Contradiction Logical Equivalence & Algebra of Proposition

Logical Equivalence:

the compound proposition p and q are logically equivalent if $p \leftrightarrow q$ is
tautology

p	q	$p \leftrightarrow q$
T	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T



$P \rightarrow q \equiv \neg P \vee q$ is logically equivalent or not.

P	q	$P \rightarrow q$	$\neg P$	$\neg P \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$P \vee q \equiv q \vee P$



Algebra of Proposition:-

1. Idempotent law:

$$P \vee P = P$$

$$P \wedge P = P$$

P	P	$P \vee P$
T	T	T
F	F	F

2. Associative law:

$$(P \vee q) \vee r \equiv P \vee (q \vee r)$$

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

3. Commutative law:

$$P \vee q \equiv q \vee P$$

4. Identity law

$$P \wedge T \equiv P$$

$$P \vee F \equiv P$$

5. double negation:

$$\neg(\neg P) \equiv P$$

$$P \equiv \neg(\neg P)$$

6. De Morgan's law:

$$\neg(P \wedge q) \equiv \neg P \vee \neg q$$

$$\neg(P \vee q) \equiv \neg P \wedge \neg q$$

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Lecture 40

Constructing new logical Equivalence Converse, Contrapositive & Inverse

Show that $\neg(P \rightarrow q)$ and $P \wedge \neg q$ are logically equivalent.

$$\begin{aligned}\neg(P \rightarrow q) &\equiv \neg(\neg P \vee q) \\ &\equiv (\neg\neg P) \wedge \neg q \quad \leftarrow \text{De Morgan law} \\ &\equiv P \wedge \neg q \quad \leftarrow \text{double negation law}\end{aligned}$$

$$\begin{aligned}&PV[\neg(\neg P \wedge q)] \text{ and } PV(\neg q) \\ &PV[\neg(\neg P) \vee (\neg q)] \quad \leftarrow \text{De Morgan law} \\ &PV[P \vee (\neg q)] \quad \leftarrow \text{Double negation law} \\ &P \vee P \vee (\neg q) \quad \leftarrow \text{Associative law} \\ &\rightarrow P \vee (\neg q) \quad \leftarrow \text{Idempotent law}\end{aligned}$$

• **Converse:** Converse of $P \rightarrow q$ is $q \rightarrow P$.

Ex The home team win whenever it is raining

- 1) if it is raining, then the home team win. $P \rightarrow q$
- 2) if the home team win, then it is raining. $q \rightarrow P$

Contrapositive of $P \rightarrow Q$.
 the Contrapositive of $P \rightarrow Q$ is $\neg Q \rightarrow \neg P$

Example:

the home team win whenever it is raining.

- if it is ^{raining} P , then the home team win ($P \rightarrow Q$)
- if the home team does not win ($\neg Q \rightarrow \neg P$) then it is not raining

• Inverse of $P \rightarrow Q$.
 Inverse of $P \rightarrow Q$ is $\neg P \rightarrow \neg Q$

Example:

- if it is raining then the home team win
- if it is not raining then the home team does not win.

$P \rightarrow Q$ / $Q \rightarrow P$. C
 $P \rightarrow Q$ / $\neg Q \rightarrow \neg P$ CP
 $P \rightarrow Q$ / $\neg P \rightarrow \neg Q$ I

Valid and Invalid Argument

Argument:-

list of statement called premises or assumption or hypotheses.

result of statements called conclusion.

		valid	invalid
P ₁	Premis	T	T
P ₂	Premis	T	T
P ₃	Premis	T	T
∴	Conclusion	T	F

Example:-

- i like math.
- math is interesting.

∴ Math is interesting that way i like it/math

Valid Argument.

if the conclusion is true
when all the premise are true

* an argument is valid if conjunction of its premises imply conclusion.

$(P_1 \wedge P_2 \wedge P_3 \dots \wedge P_n) \rightarrow C$ is a tautology

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* Invalid Argument :

if the conclusion is false
when all the premises are true.

- an argument is invalid if conjunction of its premises does not imply conclusion

$$\begin{array}{l} P \rightarrow q \\ P \\ \hline \therefore q \end{array}$$

P	q	$P \rightarrow q$	P	q
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

valid

premises		conclusion	
$P \rightarrow q$	P	$P \rightarrow q$	P
T	T	T	T
T	F	F	F

critical row $\rightarrow$ all premises are true

P	q	$P \rightarrow q$	q	P
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	F

invalid

premises conclusion

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Lecture 41

Part 2

Argument, Rule of inference for proposition logic and Example

Argument: Sequence of statement which end with conclusion

- Valid Argument.
- Invalid Argument.

Rule of inference:

$$\begin{array}{l} 1. \quad P \\ \quad P \rightarrow q \\ \hline \therefore q \end{array} \quad \begin{array}{l} (P \wedge (P \rightarrow q) \rightarrow q) \\ \text{is tautology} \end{array} \quad (\text{Modus ponens})$$

$$\begin{array}{l} 2. \quad \neg q \\ \quad P \rightarrow q \\ \hline \therefore \neg P \end{array} \quad \begin{array}{l} (\neg q \wedge (P \rightarrow q) \rightarrow \neg P) \\ \text{is a tautology} \end{array} \quad (\text{Modus tollens})$$

$$\begin{array}{l} 3. \quad P \rightarrow q \\ \quad q \rightarrow r \\ \hline \therefore P \rightarrow r \end{array} \quad (P \rightarrow q) \wedge (q \rightarrow r) \rightarrow (P \rightarrow r) \quad (\text{Hypothetical syllogism})$$

$$4.) \frac{P \vee q \quad \neg P}{\therefore q} \quad (P \vee q) \wedge (\neg P) \rightarrow q \text{ is tautology} \quad (\text{disjunctive syllogism})$$

$$5.) \frac{P}{\therefore P \vee q} \quad P \rightarrow P \vee q \text{ is tautology} \quad (\text{Addition})$$

$$6.) \frac{P \wedge q}{\therefore P} \quad P \wedge q \rightarrow P \text{ is tautology} \quad (\text{simplification})$$

$$7.) \frac{P \quad q}{\therefore P \wedge q} \quad P \wedge (q) \rightarrow P \wedge q \text{ is tautology} \quad (\text{conjunction})$$

$$8.) \frac{P \vee q \quad \neg P \vee r}{\therefore q \vee r} \quad (P \vee q) \wedge (\neg P \vee r) \rightarrow (q \vee r) \text{ is tautology} \quad (\text{Resolution})$$

Example: $P \rightarrow q$

1. if it is rainy then the pool will be closed. it is rainy therefore the pool is closed.

Let $P \rightarrow q$

P

$\therefore q$

Here modus ponens is used. so this is valid argument.

2. Ezaan is mathematic major and a Computer science major therefore Ezaan is mathematic major.

Let.

$P \wedge q$

$\therefore P$

Here simplification is used. so this is valid argument.

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Lecture 42

Propositional functions, Quantifiers and Universal Quantifier.

Propositional function:-

let A be given set. A
propositional function or open sentence $P(x)$
which has the property that $p(a)$ is true or
false for each $a \in A$

Domain: the set A is called the domain of $p(x)$

Example:

let $p(x)$ denote the statement " $x > 3$ "
What are the truth value of $p(4)$ and $p(2)$?

Sol:

Here $p(x) : x > 3$

$p(4) : 4 > 3$

is true statement

Also

$p(2) : 2 > 3$

is false statement.

Hence,

$P(4)$ is true and $P(2)$ is false.

let $p(x)$ denote the statement $x+2 > 7$.
 find truth set T_p of each proposition
 function $p(x)$ $P = \{1, 2, 3, 4, \dots\}$

Sol.

Here $P = \{1, 2, 3, 4, \dots\}$

$P(x) = \{x+2 > 7\}$

$T_p = \{6, 7, 8, 9, \dots\}$

$1+2 > 7$	$2+2$
$3 > 7$	$4 > 7$
$6+2 > 7$	$7+2 > 7$
$8 > 7$	$9 > 7$

let $Q(x, y)$ denote the statement $x = y+3$
 what are the truth value of proposition

$Q(1, 2)$ and $Q(3, 0)$

Sol.

$Q(x, y) = x = y+3$

$Q(1, 2) = 1 = 2+3$

$= 1 = 5$ false statement

$Q(3, 0) = 3 = 0+3$

$3 = 3$ true statement

• Quantifier:

it extent to which a propositional function
 is true over a range of element.

1. Universal quantification.
2. existential quantification

1. Universal Quantifier:

"For every x in A , $p(x)$ "

is true statement

$\forall x p(x)$ is true.

*the truth set of $p(x)$ is the entire set A .

Example:

let $p(x)$ be the statement " $x+1 > x$ "

What is the truth value of quantification

$\forall x p(x)$, where the domain consist of all real num?

Here $p(x) : x+1 > x$.

Because $p(x)$ is true for all real num x .

$$T_p = \{x : x \in \mathbb{R}, x+1 > x\} = \mathbb{R}$$

$$T_p = \{1, 2, 3, \dots\} = \mathbb{P}$$

So Universal quantifier $\forall x p(x)$ is true.

$$1+1 > 1$$

$$2 > 1$$

$$3+1 > 3$$

$$4 > 3$$

$$-3+1 > -3$$

$$-2 > -3$$

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Lecture 43

Existential quantifier and negation of quantified statement.

Existential Quantifier:

For some x in A $p(x)$ is true statement

$$\forall \exists x p(x)$$

the truth set of $p(x)$ is non empty may be one of two element are true then the existential quantifier is true

Example:

the proposition $(\exists x \in \mathbb{P})(x+4 < 7)$ is true?

Sol:-

$$\{n: n+4 < 7\} = \{1, 2\} \neq \emptyset$$

so existential quantifier is true ✓

$1+4 < 7$	$2+4 < 7$
$5 < 7$	$6 < 7$
$3+4 < 7$	$4+4 < 7$
$7 < 7$	$8 < 7$
	X

the proposition $(\exists n \in \mathbb{P})(n+6 < 4)$ is false?

$$\{n: n+6 < 4\} = \emptyset \quad \{ \}$$

so existential quantifier is false

$$\begin{array}{l} 1+6 < 4 \\ 7 < 4 \end{array}$$

Determine the truth value of each of these
domain of each variable is all real numbers.

1. $\exists x (x^2 = 2)$

if we put $x = \sqrt{2}$ or $x = -\sqrt{2}$

then $\exists x (x^2 = 2)$	$x^2 = 2$
$= \exists x (\sqrt{2})^2 = 2$	$(-\sqrt{2})^2 = 2$

$= 2 = 2 \neq \phi$ so existential quantifier is true.

2. $\exists x (x^2 = -1)$

if we put $x = 1$ or -1

$\exists x (1)^2 = -1$	$(-1)^2 = -1$
$1 \neq -1$	$1 \neq -1$

$\{x : x^2 = -1\} = \phi$ so existential quantifier is false

3. $\forall x (x^2 + 2 \geq 1)$

Because $T_p \{x : x^2 + 2 \geq 1\} = \mathbb{R}$

$\forall x ((1)^2 + 2 \geq 1)$	$(-1)^2 + 2 \geq 1$
--------------------------------	---------------------

$\forall x (1 + 2 \geq 1)$	$1 + 2 \geq 1$
----------------------------	----------------

$\forall x (3 \geq 1)$	$3 \geq 1$
------------------------	------------

Determine the truth value of each of these
domain of each variable is all real numbers.

1. $\exists x (x^2 = 2)$

if we put $x = \sqrt{2}$ or $x = -\sqrt{2}$

$$\begin{array}{l|l} \text{then } \exists x (x^2 = 2) & x^2 = 2 \\ = \exists x (\sqrt{2})^2 = 2 & (-\sqrt{2})^2 = 2 \end{array}$$

$= 2 = 2 \neq \phi$ so existential quantifier is true.

2. $\exists x (x^2 = -1)$

if we put $x = 1$ or -1

$$\begin{array}{l|l} \exists x (1)^2 = -1 & (-1)^2 = -1 \\ 1 \neq -1 & 1 \neq -1 \end{array}$$

$\{x : x^2 = -1\} = \phi$ so existential quantifier is false

3. $\forall x (x^2 + 2 \geq 1)$

Because $T_p \{x : x^2 + 2 \geq 1\} = \mathbb{R}$

$$\forall x ((1)^2 + 2 \geq 1) \quad | \quad (-1)^2 + 2 \geq 1$$

$$\forall x (1 + 2 \geq 1) \quad | \quad 1 + 2 \geq 1$$

$$\forall x (3 \geq 1) \quad | \quad 3 \geq 1$$

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$$\forall x (x^2 \neq x)$$

$$\forall x ((1)^2 \neq 1)$$

$$1 = 1$$

$$(4)^2 \neq 4$$

$$16 \neq 4$$

so universal quantifier is false.

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Lecture 43

Negation of Quantified Statement

part 2

• Negation of Universal Quantifier: $\neg$

• "x has taken a course in calculus" domain consist of all student in your class

• "Every student in your class has taken a course in calculus"

$$\forall x p(x)$$

universal quantifier "if it is not the case that every student in your class has taken in calculus"

$$\exists x \neg p(x)$$

$$\neg \forall x p(x) \equiv \exists x \neg p(x)$$

• Negation of Existential Quantifier:-

• "There is a student in this class, who has not taken a course in calculus"

$\neg$ "there is no student in class who has not taken a course in calculus"

"Every student in class has not taken a course in calculus."

$$\forall x \neg p(x)$$

$$\Rightarrow \neg \exists x p(x) \equiv \forall x \neg p(x)$$

Both are called de Morgan Law.

Examples:

negation of $\forall x (x^2 > x)$

$$\neg \forall x (x^2 > x) \equiv \exists x \neg (x^2 > x)$$

$$\equiv \exists x (x^2 < x)$$

negation of $\exists x (x^2=2)$

$$\neg \exists x (x^2=2) \equiv \forall x \neg (x^2=2) \\ \equiv \forall x (x^2 \neq 2)$$

- Universal quantifier $\equiv$ ^{domain} conjunction $\wedge$
- Existential quantifier $\equiv$ disjunction $\vee$

Q domain is $\{1, 2, 3, 4\}$ write out each using disjunction, conjunction and negation.

- $\exists x p(x) \equiv P(1) \vee P(2) \vee P(3) \vee P(4)$

- $\forall x p(x) \equiv P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

- $\exists x \neg p(x) \equiv \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

- $\forall x \neg p(x) \equiv \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

- $\neg \exists x p(x) \equiv \forall x \neg p(x) \equiv \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

$$\neg \forall x p(x) \equiv \exists x \neg p(x) \equiv \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$$

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Lecture 44

Introduction to proof and method of Proving theorem Part 1.

• Introduction:

Before proof we have to get to know some terminology.

✓ theorem:

A theorem is a statement that can be shown to be true.

• Proposition:

less important theorem are called proposition.

• Proof:

A proof is valid argument that establish the truth of a theorem.

i. Axiom:

Statement that are assume to be true. used in proof.

• Lemma:

less important theorem use to proof other theorem

Corollary:

is a theorem that can be established directly from a theorem that has been proof.

• Conjecture:

statement that is being proposed to be true statement, usually base on some partial evidence.

• Method of Proving Part 1:
technique:

1. Direct Proof
2. Indirect proof

• Direct Proof:

A direct proof of conditional statement $p \rightarrow q$ is constructed when the first is assumption that p is true. subsequent step are constructed using axiom, definition and previously proved theorem together with rule of inference to show that q must also be true.

Even odd Integer.

The integer $n = 2k$ where n is even
and if n is odd $n = 2k+1$

Example.

even $n = 6 = 2(3)$

odd $n = 5 = 2(2)+1$

Q Give a direct proof of the theorem
"if n is an odd integer, then n^2 is odd"

Sol:-

Let P be the statement that n is an odd integer and q be the statement that n^2 is an odd integer

Assume that p is true.

by definition $n = 2k+1$

then

$$n^2 = (2k+1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

$$\text{put } L = 2k^2 + 2k$$

$$n^2 = 2L + 1$$

Hence n^2 is odd.

$$\begin{aligned} & (2k)^2 + 2(2k)(1) + (1)^2 \\ & 4k^2 + 4k + 1 \end{aligned}$$

$$n = 2k+1$$

$$n^2 = 2L+1$$

Q. Give a direct proof that if m and n are both perfect square then nm is also a perfect square.

Sol:-

Assume that m and n are perfect square.

then

$$m = r^2$$

$$n = s^2$$

$$4 = (2)^2$$

$$16 = (4)^2$$

$$\begin{aligned} \text{Now } m \cdot n &= r^2 \cdot s^2 \\ &= (rs)^2 \end{aligned}$$

$$mn = t^2$$

$$t = rs$$

Hence proved.

Prove that n is any even integer then $(-1)^n = 1$

Assume that n is any even integer then $n = 2k$

$$\begin{aligned} (-1)^n &= (-1)^{2k} \\ &= [(-1)^2]^k \\ &= 1^k = 1 \end{aligned}$$

$$-1 \times -1 = 1 \quad 1^2 \mid 3$$

Hence prove.

Use direct proof to show that the product of two rational num is rational.

Sol:-

$$\text{let } x = \frac{a}{b}, b \neq 0$$

$$y = \frac{c}{d}, d \neq 0$$

$$\text{Now: } x \cdot y = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

$$ac = m, bd = n$$

$$xy = \frac{m}{n}, n \neq 0$$

Hence prove,

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Lecture 45

Method of proving theorem part 2

Indirect Proof:-

the proof that do not start with the premises and end with conclusion.

1. Proof by Contraposition.
2. Proof by Contradiction.

$$P \rightarrow Q$$

Proof by Contraposition:

$$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$$

Examples:-

Prove that if (n is an integer and $3n+2$ is odd) then (n is odd.)

Sol:- Direct Proof

Suppose n is odd is true

$$3n+2 = 2k+1$$

$$3n+1 = 2k$$

$$n = \frac{2k-1}{3}$$

Proof by Contraposition.

Suppose n is odd is false.

$$n = 2k$$

$$3n+2 = 3(2k)+2$$

$$3n+2 = 6k+2$$

$$3n+2 = 2(3k+1) \text{ therefore } n \text{ is even}$$

$$\begin{aligned} 3n &= 6k+2 \\ 3n &= 6k \\ n &= 2k \end{aligned}$$

Prove that if $n = ab$ where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Sol:-

Direct Proof

Suppose $n \neq ab$ is true.
 $n = ab$

Because there is no obvious way to show that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ so direct method is not working

Proof by Contraposition
Suppose $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$ is false.

$a \leq \sqrt{n} \vee b \leq \sqrt{n}$ is false

this implies that

$a > \sqrt{n}$ and $b > \sqrt{n}$

$ab > (\sqrt{n})^2$

$ab > n \quad \text{uq} \rightarrow \text{up}$

$ab \neq n$

Hence proof by contraposition. Succeed

Prove that if n is an integer and n^2 is odd then n is odd.

* Direct Method:

Suppose n is odd integer and n^2 is odd

$$n^2 = 2k+1$$

$$\sqrt{n^2} = \sqrt{2k+1}$$

$$n = \sqrt{2k+1}$$

* Proof by contrapositive:

Suppose n is odd is false
then,

$$n = 2k$$

$$n^2 = (2k)^2$$

$$n^2 = 4k$$

$$n^2 = 2(2k)$$

n^2 is even then proof that
over statement is true.

Proof by Contradiction.

Give a proof by Contradiction of theorem. if $3n+2$ is odd then n is odd

Sol:

Suppose p , $3n+2$ is odd
 q , n is odd

$$\neg q \rightarrow \neg p$$

$$p \rightarrow \neg q$$

Assume p and $\neg q$ are true.

1) $3n+2$ is odd

2) n is even

$$n = 2k$$

$$3n+2 = 3(2k)+2$$

$$= 6k+2$$

$$3n+2 = 2(3k+1)$$

$$3n = 6k+2-2$$

$$3n = 6k$$

$$n = \frac{6k}{3}$$

$$n = 2k$$

$$\neg p \quad 3n+2 \text{ is even}$$

Because p and $\neg p$ are true
we have contradiction.

1. Direct proof: $P \rightarrow Q$

Suppose P is true then Q is true

2. Proof by Contraposition: $P \rightarrow \neg Q$

Suppose Q is false $\neg Q \rightarrow$

3. Proof by Contradiction:

Suppose P and $\neg Q$ is true.